



**LOCOMOTIVE SHED, SABARMATI
ELECTRICAL**

IA/IB/IC Schedule form for KAVACH (TICAS) in WAG9HC/WAP7 Locomotive



Loco No.: _____

Sch.: _____

Date: _____

On Locomotive

Sr.	Description of the Equipment	Parameter for quality check	Details of work done		Remarks	IA/IB	IC
			Cab 1	Cab 2			
1.1.1	LPOCIP	Cleanliness				✓	✓
1.1.2		Physical condition				✓	✓
1.1.3		Fixing condition				✓	✓
1.1.4		Check the intactness of couplers					
1.1.5		Open all couplers and check all its pins					
1.1.6		Ensure working of Horn from DMI				✓	✓
1.2.1	Cab Radio Display Unit (If applicable)	Cleanliness				✓	✓
1.2.2		Physical condition				✓	✓
1.2.3		Fixing condition				✓	✓
1.2.4		Check the intactness of couplers					
1.2.5		Open all couplers and check all its pins					
2.1.1	Buzzer	Cleanliness				✓	✓
2.1.2		Physical condition					
3.1.1	Radio Unit	Cleanliness				✓	✓
3.1.2		Physical condition				✓	✓
3.1.3		Intactness of Connectors and Couplers				✓	✓
3.1.4		Fixing condition				✓	✓
4.1.1	LTE Router-1 (If applicable)	Cleanliness				✓	✓
4.1.2		Physical condition				✓	✓
4.1.3		Intactness of Connectors and Couplers				✓	✓
4.1.4		Fixing condition				✓	✓
5.1.1	LTE Router-2 (If applicable)	Cleanliness				✓	✓
5.1.2		Physical condition				✓	✓
5.1.3		Intactness of Connectors and Couplers				✓	✓
5.1.4		Fixing condition				✓	✓

Under frame:

Sr.	Description of the Equipment	Parameter for quality check	Details of work done		Remarks	IA/IB	IC
			Cab 1	Cab 2			
7.1.1	RFID Reader	Cable Conduit clamping condition				✓	✓
7.1.2		RFID Reader Mounting fixture welding Intactness (visual check)				✓	✓
7.1.3		Ensure intactness of welding by Magnetic particle testing/DPT					✓
7.1.4		Availability and intactness of Sling for RFID Reader				✓	✓
7.1.5		Intactness of Guard for RFID reader (wherever available)				✓	✓

On Locomotive (As per availability)

Sr.	Description of the Equipment	Parameter for quality check	Details of work done (only one item / locomotive)	Remarks	IA/IB	IC
8.1.1	Onboard KAVACH	Cleanliness			✓	✓
8.1.2		Physical condition			✓	✓
8.1.3		Intactness of Connectors and Couplers			✓	✓
8.1.4		Fixing condition			✓	✓
8.1.5		Feeding/ensuring of correct wheel dia. As per the wheel dia. Measurement carried out by Shed. (Whenever unscheduled wheel turning done also wheel dia. Feeding need to be ensured). Data feeding need to be done by KAVACH OEM. The updated wheel dia. Must be reflected Either in configuration tool or DMI.			✓	✓
8.1.6		Intactness of all connectors/couplers			✓	✓
8.1.7		Open all couplers and check all its pins				
8.2.1	Counter Readings	Note Isolation counter reading			✓	✓
8.3.1	CAB Input Box (If Applicable)	Cleanliness			✓	✓
8.3.2		Physical condition			✓	✓
8.3.3		Intactness of Connectors and Couplers			✓	✓
8.3.4		Fixing condition			✓	✓
8.4.1	E70/CCB Interface Unit(If applicable)(HBL)	Cleanliness			✓	✓
8.4.2		Physical condition			✓	✓
8.4.3		Intactness of Connectors and Couplers			✓	✓
8.4.4		Fixing condition			✓	✓
8.5.1	MCBs (2 nos.)	Check Working and Tripping of MCBs on Test Bench				
9.1.1	LE Valve (E-70)	Fixing condition				
9.1.2		Overhauling of LE Unit				
9.1.3		Measurement of IR Value of coil				
10.1.1	Isolation Box	If Isolation switch kept in Isolation position whether LPOCIP is indicating the status or not			✓	✓
10.1.2		If Isolation switch kept in Isolation position whether traction and braking effort are coming or not			✓	✓
10.1.3		Cleanliness			✓	✓
10.1.4		Physical condition			✓	✓
11.1	Addl. Q51 (HBL)	Cleanliness			✓	✓
11.2		Physical condition			✓	✓
11.3		Connections/Contacts				✓

Under frame:

Sr.	Description of the Equipment	Parameter for quality check	Details of work done (only one item / locomotive)	Remarks	IA/IB	IC
12.1.1	Speed Sensor I	Visual check of Speed Sensor Cables			✓	✓
12.1.2		Visual Inspection of PG mounting and Junction box			✓	✓
12.1.3		Remove the speed sensor and clean any excess grease (Applicable for conventional locos)				✓

12.1.4		Ensure checking of fork bolt/drive pin				✓
12.2.1	Speed Sensor II	Visual check of Speed Sensor Cables			✓	✓
12.2.2		Visual Inspection of PG mounting and Junction box			✓	✓
12.2.3		Remove the speed sensor and clean any excess grease (Applicable for conventional locos)				✓
12.2.4		Ensure checking of fork bolt/drive pin				✓

Roof

Sr.	Description of the equipment's	Parameter to be checked:	Details of work done	Remarks:	IA/IB	IC
13.1.1	GSM/GNSS (GPS)	Physical condition			✓	✓
13.1.2	Antenna on roof (In some makes only one unit available)	Cable conduit clamping condition			✓	✓
			Antenna-1	Antenna-2		
14.1.1	Antennae on Roof of CAB1	Physical condition			✓	✓
14.1.2		Cable Conduit clamping condition			✓	✓
14.1.3		Checking Reverse Power status (less than 1 Watt)			✓	✓
14.1.4		Ensure of intactness of welding by RDPT				
14.2.1	Antennae on Roof of CAB2	Physical condition			✓	✓
14.2.2		Cable Conduit clamping condition			✓	✓
14.2.3		Checking Reverse Power status (less than 1Watt)			✓	✓
14.2.4		Ensure of intactness of welding by RDPT				

LTE Antenna – 1 & 2

Sr.	Description of the equipment's	Parameter to be checked:	Details of work done	Remarks:	IA/IB	IC
14.3.1	LTE Antenna-1 (If applicable and not available in KAVACH 4.0)	Physical condition			✓	✓
14.3.2		Cable Conduit clamping condition			✓	✓
14.4.1	LTE Antenna-2 (If applicable and not available in KAVACH 4.0)	Physical condition			✓	✓
14.4.2		Cable Conduit clamping condition			✓	✓
15.1	Data downloading from On-board KAVACH	Data downloading for analyzing			✓	✓

System Functionality:

Note: As per requirement whenever any KAVACH related attention paid or based on the log-book remarks. This in-line with Railway Board guidelines: 2025/Elect. (TRS)/440/5- Part (1) dated 06.11.2025 and Railway Board Letter no.2025/Elect (TRS)/440/5 (KAVACH) DT. 30.03.26

Sr.	Description of the equipment's	Parameter to be checked:	Details of work done	Remarks:	IA/IB	IC
15.2	System Functionality	Testing as per the Station KAVACH Layout in Shed (in Trip sheds subject to availability of Station KAVACH)			✓	✓
15.3		Intactness of Checksum of Loco KAVACH (This in-line with Railway board guidelines 2025 / Elect. (TRS) / 440 /5-Part(1) dated 06.11.25)			✓	✓

Sr.	Description of the equipment's	Parameter to be checked:	Details of work done	Remarks:	IA/IB	IC
15.4		Self-testing to ensure healthiness and working of system			✓	✓
		Ensure pressure drop during self-test.			✓	✓
		Checksum to be verified			✓	✓
		Tightness of foundation bolts			✓	✓
		Scenario testing on test track, systems action on Loss of RF communication and Introduction of Foreign tag on the test track may be check.			✓	✓
		DMI analysis and brake testing shall be made mandatory for minor and major schedules, as detailed below:				
		Radio-1,2 Diag. status				
		RFID-1,2 link: OK				
		TX/RX power: should be in limit				
		DMI-1,2 Link: OK				
		EVL: OK				
		(vi) GPS-1 and GPS-2 should be check Pneumatic testing should be done after major schedule				

KAVACH Specific Testing proforma during IA/IB/IC and Major Schedules:

Sr.	Parameter to be checked	Status of work	Remarks
1.	When KAVACH is Off: Ensure availability of TE and BE from each Cab. For conventional locomotives, ensure availability of Notch progression in traction and dynamic braking from each Cab.		
2.	When KAVACH is Switched On but Isolation Switch is kept in Isolation Position: Ensure availability of TE and BE from each Cab. For conventional locomotives, ensure availability of Notch progression in traction and dynamic braking from each Cab.		
3.	Now operate the cocks and Isolation Switch to Keep KAVACH in working condition. Kavach performs brake testing and on successful completion of brake testing, enter train configuration as Light Engine. Now ensure availability of TE and BE from each Cab. For conventional locomotives, ensure availability of Notch progression in traction and dynamic braking from each Cab.		
4.	Check the indications in DMI This ensures correct integration of Kavach with brake system and traction cut off interfaces		

Driver Bookings & attention given:

Any other remarks:

Sign of TCN:

Sign of JE/SSE: